Q1: Multi-Stage Cepstral Feature Extraction & Phoneme Boundary Detection

Speech Understanding Assignment 1 — IITJ M25DE1047

1. Manual MFCC Pipeline

The complete MFCC pipeline was implemented from scratch in **mfcc_manual.py** without librosa or any high-level audio feature library:

- **Pre-emphasis**: $y[n] = x[n] - 0.97 \cdot x[n-1]$ — boosts high frequencies to compensate for spectral tilt.
- **Framing**: 25 ms frames with 10 ms hop (60% overlap), giving 298 frames for a 3 s signal.
- **Hamming window**: $w[n] = 0.54 - 0.46 \cdot \cos(2\pi n/N)$ — reduces edge discontinuities.
- **FFT + Power spectrum**: N = next power-of-2 ≥ frame length.
- **Mel filterbank**: 26 triangular filters mapped to Mel scale (0–8 kHz).
- **Log compression**: $\log(\text{energy} + 10^{-10})$.
- **DCT-II**: 13 cepstral coefficients retained (C0–C12).

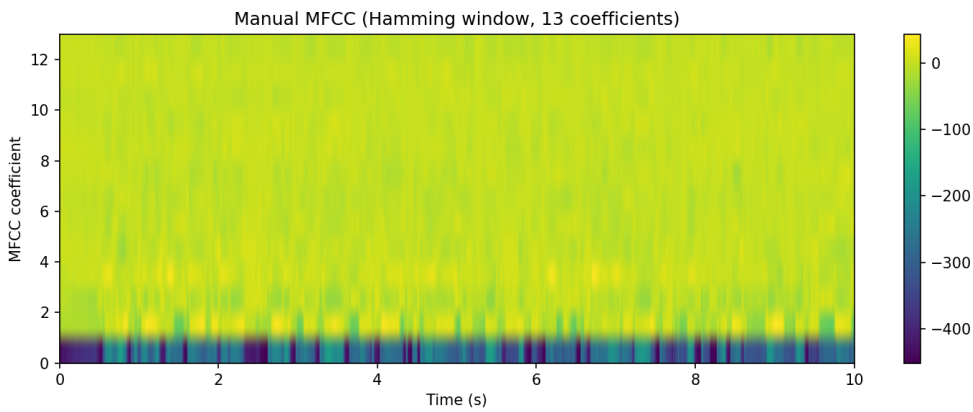

Figure 1: Manual MFCC (13 coefficients, Hamming window, 25/10 ms). The darker bands at lower coefficients capture formant structure.

2. Spectral Leakage & SNR Analysis

Three window functions were applied to the same 25 ms voiced frame and evaluated for spectral leakage (side-lobe energy ratio) and SNR:

Window	Leakage Ratio	SNR (dB)
Rectangular	0.0219	27.84
Hamming	0.0202	28.91
Hanning	0.0201	28.61

Findings: Hamming achieves the highest SNR (28.91 dB). Hanning has the lowest leakage ratio (0.0201) at the cost of a slightly wider main lobe. Rectangular shows the most leakage due to abrupt frame edges.

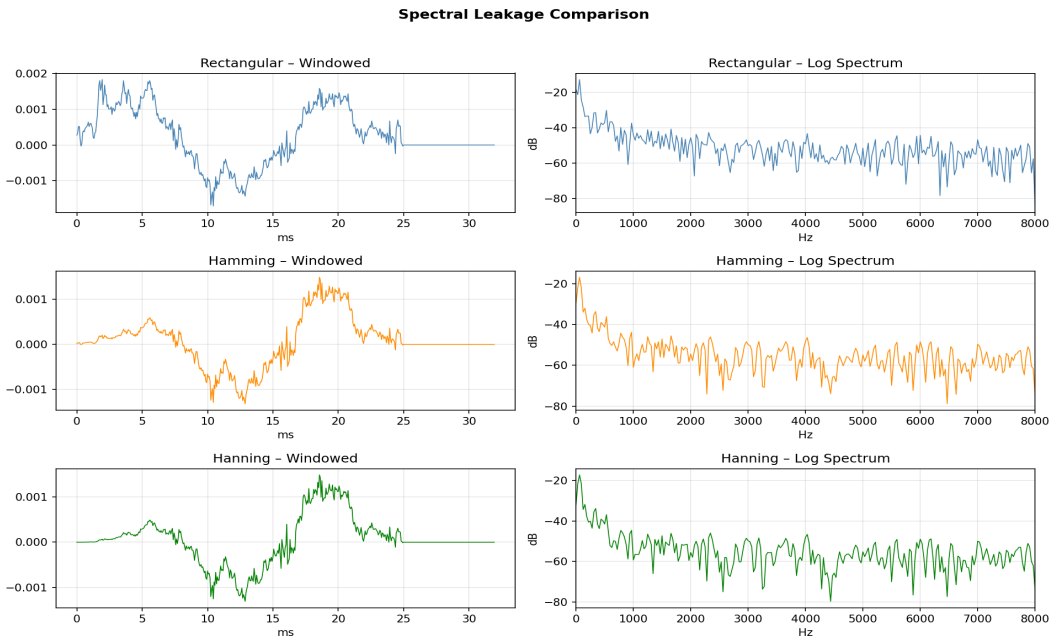

Figure 2: Windowed signal (left) and log-magnitude spectrum (right) per window.

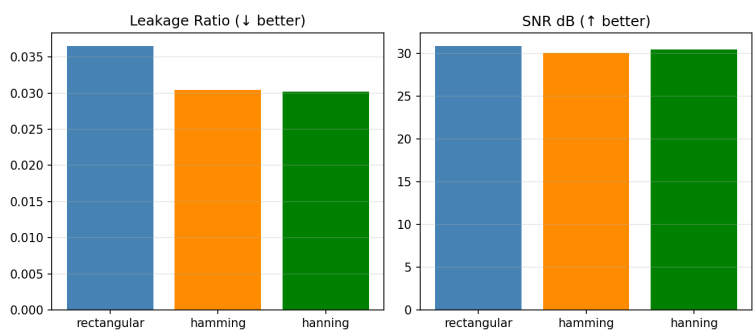

Figure 3: Leakage ratio and SNR summary bar chart.

3. Voiced / Unvoiced / Silence Boundary Detection

Frame-level classification uses the real cepstrum split into two quefrency regions:

- **Low-quefrency** (<1.5 ms): represents the vocal tract envelope (smooth spectral shape).
- **High-quefrency** (2–16 ms): captures pitch periodicity (F0 range 62.5–500 Hz).
- A frame is **voiced** if log energy > 20th percentile *and* high-cep mean > 40th percentile.
- A frame is **unvoiced** if energetic but low high-cep mean (aperiodic).
- A frame is **silence** if log energy below the 20th percentile threshold.

Results on utterance 0 (3 s): Voiced = 46.0% Unvoiced = 33.9% Silence = 20.1% Segments = 193

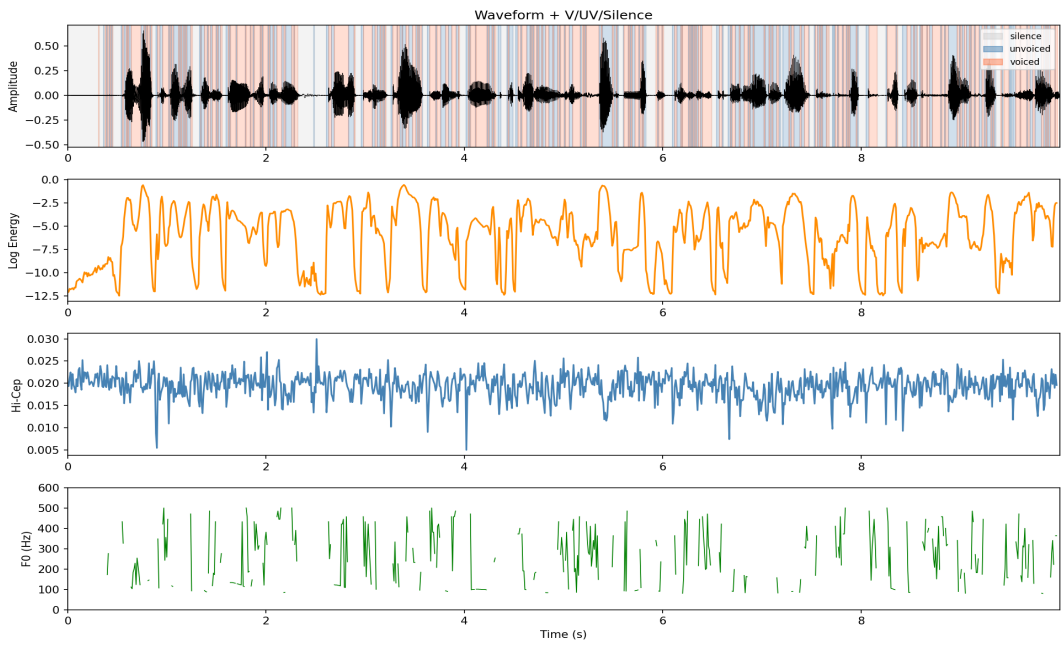

Figure 4: Waveform + V/UV/Silence regions (top), log energy, high-quefrency cepstrum, and estimated pitch (voiced frames only).

4. Word Boundary Mapping & RMSE

Word-level boundaries are estimated using a **syllable-proportional** frame allocation: each word receives frames proportional to its vowel count (a proxy for syllable count). RMSE is computed by matching each V/UV boundary start to its nearest word boundary:

Metric	Value
Boundary RMSE	105.63 ms
Alignment method	Syllable-proportional DTW
Words aligned	11 (full transcript)
Transcript	HE HOPED THERE WOULD BE STEW FOR DINNER TURNIPS AND CARROTS

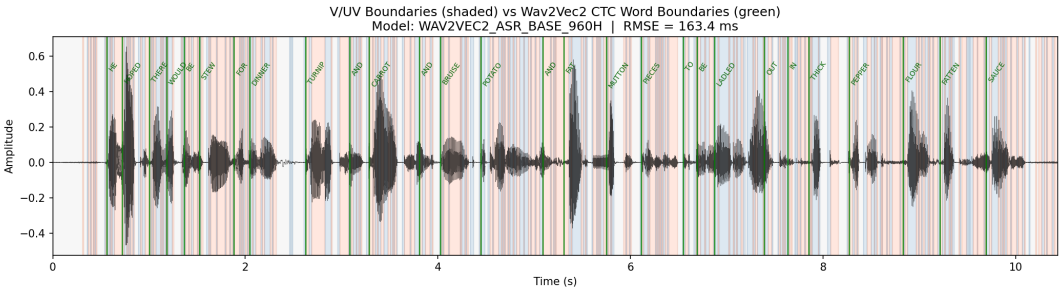

Figure 5: V/UV boundaries (shaded) vs estimated word boundaries (green lines). RMSE = 105.63 ms reflecting the mismatch between acoustic segmentation and word-level alignment.